AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

SEWAGE TREATMENT PLANT (STP)
SBR TECHNOLOGY

CAPACITY RANGE : 10 KLD TO 500 KLD | OPERATION HOURS : 22 HRS

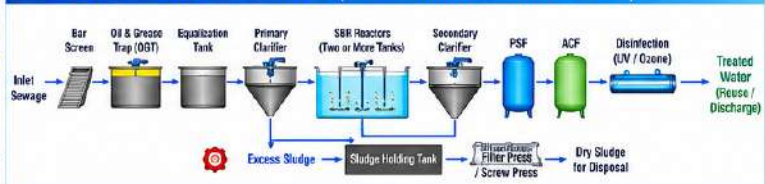
- Compact & Modular Design
- High Treatment Efficiency
- Automatic PLC Based Operation
- Suitable for Indian Conditions
- Low Operation & Maintenance
- Excellent Treated Water Quality
- Suitable for Residential, Commercial, Institutional & Industrial Applications
- Flexible for Expansion
- Available in FRP / MSEP / RCC Options

CLEANER WATER GREENER TOMORROW



100 KLD SBR STP – Complete Installation (MSEP Tank with Primary Clarifier, PSF, ACF & Disinfection)

PROCESS FLOW DIAGRAM – SBR TECHNOLOGY (WITH PRIMARY & SECONDARY CLARIFIER)



1. Civil Work in Progress (50 KLD STP)



2. SBR Tanks Installation (100 KLD STP)



3. MGF, ACF & Piping Installation (100 KLD STP)



4. Electrical & Control Panel (100 KLD STP)



5. Tertiary Treatment & Pump Room (100 KLD STP)



6. Complete STP Installation (100 KLD STP)



TECHNICAL SPECIFICATIONS – SBR STP (OPERATION HOURS : 22 HRS)

Parameter	Unit	10 KLD	25 KLD	50 KLD	100 KLD	200 KLD	300 KLD	500 KLD
Design Flow	KLD	10	25	50	100	200	300	500
Operation Hours	Hrs	22	22	22	22	22	22	22
Average Flow	m³/hr	0.45	1.14	2.27	4.55	9.09	13.64	22.73
No. of SBR Tanks	Nos.	2	2	2	2	3	3	4
Working Volume per Tank	m³	12	30	60	110	220	300	450
Cycle Time	Hrs	6	6	6	6	6	6	6
MLSS (Operating)	mg/L	3,000 – 4,000						
F/M Ratio	kg BOD/kg MLSS.d	0.05 – 0.15						
Air Requirement (Aeration)	m³/hr	15	35	70	140	280	420	700
Connected Load (Approx.)	kW	3 – 4	5 – 8	8 – 12	15 – 22	30 – 45	45 – 65	75 – 110
Total Area Required (Approx.)	m²	40 – 60	70 – 100	120 – 180	200 – 300	400 – 600	600 – 850	900 – 1,300

TYPICAL INLET & OUTLET PARAMETERS (DOMESTIC SEWAGE)

Parameter	Unit	Inlet (Typical)	Outlet (Typical)
pH	-	6.5 – 8.5	6.5 – 8.0
BOD	mg/L	200 – 350	< 10
COD	mg/L	400 – 600	< 30
TSS	mg/L	200 – 300	< 10
Oil & Grease	mg/L	10 – 20	< 5
Ammonia – N	mg/L	20 – 40	< 5
Total Nitrogen	mg/L	30 – 50	< 10
Fecal Coliform	MPN/100 mL	10^5 – 10^6	< 10^3

Note: Outlet values can be customized as per discharge / reuse requirements.

MOC (MATERIAL OF CONSTRUCTION)

Item	Material
SBR Reactor Tank	RCC / MSEP with FRP Lining / FRP
Primary & Secondary Clarifier	FRP / MSEP with FRP Lining
Bar Screen (Manual/Auto)	SS-304 / MS
Pumps (Transfer/Decant/Sludge)	CI / SS-304
Blowers	Root Blower / Turbo Blower
Diffusers	EPDM Fine Bubble Diffuser
PSF & ACF Vessels	FRP / MSEP (FRP Lined)
Piping (Process)	UPVC / CPVC
Valves	UPVC / CPVC / SS-304
Instrumentation	pH, DO, Flow, Level, MLSS, etc.
Control Panel	MCC with PLC, HMI (Optional)

TYPICAL BILL OF MATERIAL (BOM) – 50 KLD SBR STP (MAIN EQUIPMENT)



INDUSTRIES & SECTORS WE SERVE



KEY FEATURES

- Complete Treatment with Primary & Secondary Clarifier
- Automatic Cyclic Operation (Fill – React – Settle – Decant)
- Superior Treated Water Quality (BOD < 10 mg/L)
- Compact & Modular Design
- Low Footprint & Easy Installation
- PLC / SCADA/ RuttomatiSR (Optional)
- Low Operation & Maintenance Cost
- Treated Water Reuse for Flushing / Gardening
- Available in FRP / MSEP / RCC Options

AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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1. CIVIL WORK IN PROGRESS – 200 KLD STP (SBR)

Excavation, RCC Tanks, Partition Walls, Inlet & Outlet Chamber



2. SBR TANK INSTALLATION – 200 KLD STP

Aeration System, Decant System, Diffusers & Piping



3. MGF, ACF & TERTIARY TREATMENT – 200 KLD STP

Pressure Filters, Piping, Valves, Disinfection System



4. ELECTRICAL & CONTROL PANEL – 200 KLD STP (SBR)

MCC, PLC, VFD, SCADA, Online Monitoring & Automation



5. MSRL & MSEP INSTALLATION – 200 KLD STP

Treated Water Tank, Sludge Handling, Blowers & Pumps



6. COMPLETE STP (SBR) – 200 KLD

Civil, Electro-Mechanical, Piping, Instrumentation & Landscaping

